

Class Test-1

September 2025

B.Tech - 1st SEMESTER (CE12)

Basic Electrical Technology (ESC-101-A)

Time: 90 mins

Max. Marks:15

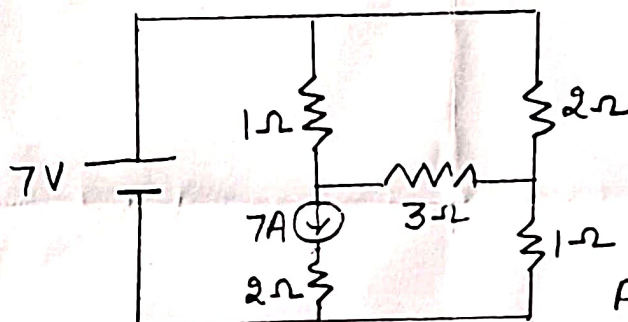
Instructions: Attempt all questions.

PART-A

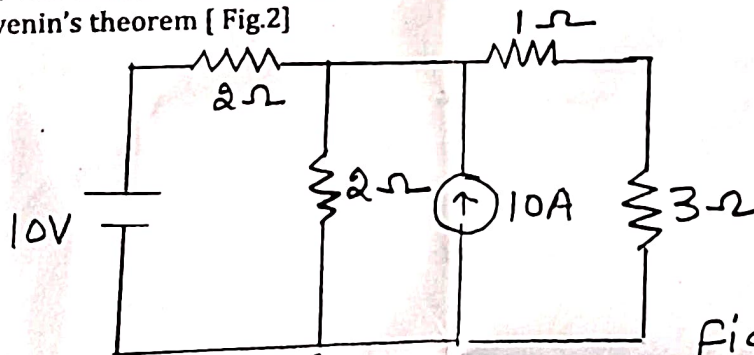
- Q1 (a) What are different types of dependent sources? (1)[C01]L1
- ✓(b) Enumerate various advantages of AC system over DC systems. (1)[C02]L1
- ✓(c) How much efficiency is obtained under the condition of maximum power transfer. List the various applications of maximum power transfer theorem. (1)[C01]L1
- ✓(d) Define the terms (i) phase and (ii) phase difference in case of alternating quantity. (1)[C02]L1

PART-B

- Q2 Find the value of current flowing through 3Ω resistor using Mesh Analysis (3)[C01]L2 for the circuit shown in Fig.1



- Q3 State Thevenin's theorem. Find the current through 3Ω resistor by using Thevenin's theorem [Fig.2] (3)[C01]L2



- Q4 Show that in a purely inductive circuit current lags the voltage by 90° . Also show that the average power consumed by it is always zero. (2.5) [C02]L2

- (b) For a half wave rectified alternating current, find the (i) Average value (ii) r.m.s value and (iii) Form factor. (2.5) [C02]L2

$\frac{P_m}{A_m}$

$$\left\{ \frac{27}{\pi} \right\}$$

$$\left\{ \frac{2}{2\sqrt{2}} \right\}$$